

Chedam — Master Specification v1.0

Oct 2, 2026 · @sreya brahma teja

Chedam is an offline-first business system — POS, inventory, team and back office — that a small store runs on its own hub over its own Wi-Fi, with no subscription and no internet dependency. This master specification combines v0.1, v0.2, v0.3 and the gap-list decisions and is the single reference for every development phase.

0. Document control

Item	Value
Document	Chedam Master Specification
Version	1.0 (baseline for development)
Owner	Sreya (product owner, support lead)
Supersedes	v0.1 Product & Technical Specification, v0.2 Change Requests, v0.3 Decisions and New Features, Reality Check and Gap List decisions
Change rule	Any change after this baseline is recorded in the Decision log (Section 15) and bumps the version (1.1, 1.2...)

How to use this document

- **Requirement IDs** (e.g. FR-3.04) are stable: code, tests and tasks refer to them.
- **Priority** uses MoSCoW: **M** must have for the phase, **S** should have, **C** could have, **L** later (planned beyond Phase 5).
- **Phase** refers to the release plan in Section 13 (P0–P5, Premium, Later).
- **Module** says which switchable module a requirement belongs to (Section 6); Core requirements are always on.
- When this document and an earlier version disagree, this document wins.

1. Executive summary

Small stores pay monthly fees and card-processing margins to POS companies, depend on internet to sell, and get little help with perishables. Chedam runs the whole store — selling, stock, staff, suppliers and books — on a low-cost hub the store owns, over its own

Wi-Fi, and keeps selling when the internet is down.

- **Offline-first:** a Raspberry Pi hub runs the app and database; tills, PCs and phones connect over the shop network. Internet is optional and used only for card-terminal linking, owner remote view, email fallback and updates.
- **One product, many stores:** each store configures name, logo, people, taxes and modules in a setup wizard; features are switched on per store to keep it simple.
- **Strengths:** perishables (lots, expiry, near-expiry markdowns), packs and singles, phone barcode receiving, bill-photo receiving, promotions and labels, loyalty without contact details, team messaging and device pings.
- **Money:** cash handled fully; cards taken on the store's own terminal (Square or similar), linked directly in a later phase.
- **Delivery:** six phases (P0–P5) from foundation to connected services, piloted in a real store before sale; a premium tier adds AI and camera features on stronger hardware.

2. Business context, objectives and success measures

Problem

- Small retailers rely on cloud POS products that charge monthly add-ons and processing fees and hold their data.
- Internet outages stop sales or reports on many systems.
- Perishables (lots, expiry, storage, wastage) are poorly handled by general POS tools.
- Owners juggle separate tools for staff messages, schedules, expenses and supplier bills.

Objectives

#	Objective	Measure of success
O1	No recurring software fee	\$0/month software cost to the store
O2	Sell without internet	100% of cash sales and stock updates work with internet off
O3	Never lose a sale or record	0 lost or duplicated sales in pilot; nightly verified backups
O4	Fast onboarding	Store live in 1 day; staff trained in under 30 minutes on the till

#	Objective	Measure of success
O5	Accurate stock	Pilot stock count variance under 2% of value after 2 months
O6	Less waste	Measurable fall in expired-stock value after near-expiry markdowns go live
O7	Easy switching	90%+ of a store's product file imported without manual retyping
O8	Product business	First paying store after a 2-month pilot

Positioning

"Your POS, your data, your hardware. No monthly fee. Keeps selling when the internet is down. Built for stores that sell perishables."

3. Scope, assumptions and constraints

In scope (Phases 0–5)

Setup and configuration; users, roles and devices; dashboard and tasks; messages, pings, notifications; checkout and payments recording; tax, returns and exchanges; products, pricing, promotions, labels; inventory, receiving, packs; customers and loyalty; vendors, purchasing, delivery-app orders; expenses; time, schedules, HR and payroll preparation; finance, reconciliation and reports; import/export and bundled datasets; connected services (terminal link, remote view, email fallback, updates).

Out of scope for now

Gift cards; email/SMS receipts; marketing campaigns and consent tracking; CCTV integration; multi-language interface; e-commerce storefront (Later); multi-location sync (Later); automatic government filings; automatic payroll deductions calculation (Later, via payroll service).

Premium tier (separate hardware)

Camera cash-count estimate; premium bill reading; snap-the-ticket for delivery orders; offline AI features.

Assumptions

- Single store location per hub; 2–10 devices in the pilot.

- First market British Columbia, Canada (CAD, GST 5% + PST 7%), designed for other provinces via tax tables.
- Staff devices run a current browser (Chrome, Edge, Safari, Firefox).
- The store owns a Wi-Fi router; internet may be absent or intermittent.
- Sreya provides installation and support during pilot and early sales.

Constraints

- \$0 recurring cost for the store's software; free/open-source components only.
- Pilot hub is a Raspberry Pi Zero 2 W (512 MB RAM): heavy work runs in browsers, not on the hub.
- Browsers require trusted HTTPS for camera and offline install.
- Card processing and delivery platforms require internet and partner agreements outside Chedam's control.

4. Stakeholders, users and roles

Stakeholders

Stakeholder	Interest	Involvement
Chedam product owner (Sreya)	Product vision, sales, support	Decides scope; signs off each phase
Store owner	Profit, control, low cost, no data loss	Pilot user; sets up store; approves money actions
Store manager	Smooth daily operations	Main daily back-office user
Cashier / staff	Fast, simple screens	Till, stock, time clock users
Store accountant	Clean records, tax reports	Read and export finance data
Vendors	Orders, bills, payments	Records only; no login
Store customers	Fast checkout, fair returns, loyalty	Indirect; loyalty and feedback
Payment and delivery partners	Integration rules	External dependencies

Roles

Role	Summary
Owner	Full control; logs in on any device; manages all devices; sends to all devices; switches modules; edits everyone's access
Manager	Runs daily operations; approves refunds, expenses, time sheets; edits access of non-owners up to their own level
Cashier	Sells, returns within limits, opens/closes till
Staff	Receives stock, counts, labels, tasks
Accountant	Reads finance and reports, exports, reconciles

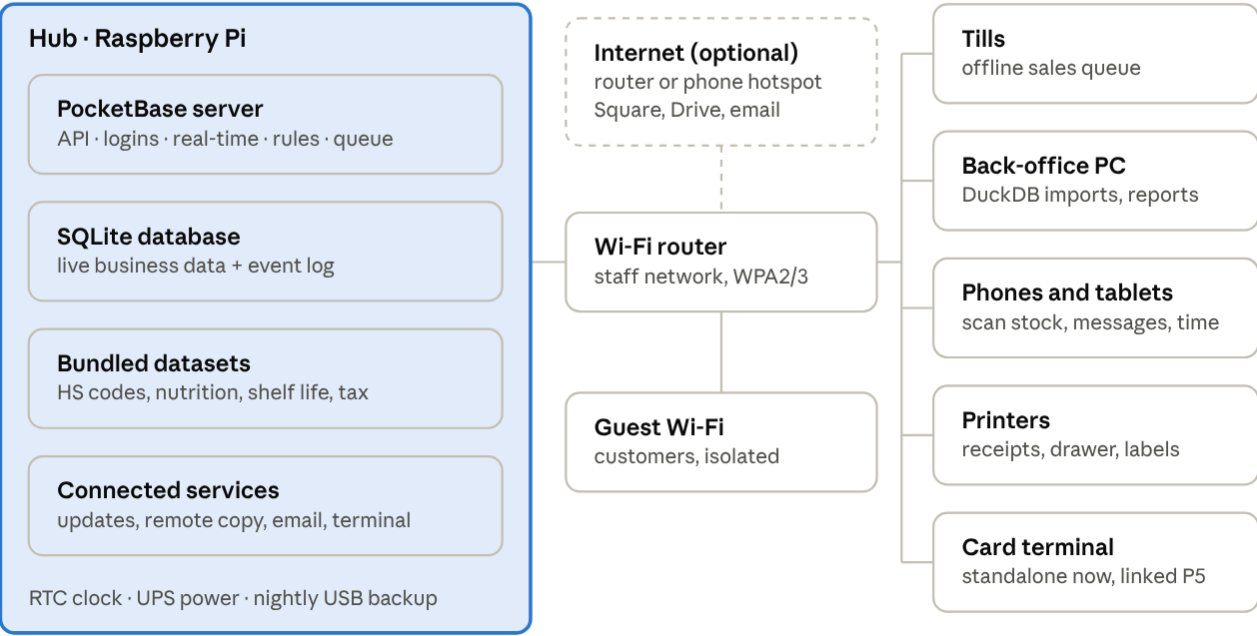
Access principles

- Roles are templates; each person can have **overrides** (extra or removed permissions), optionally **temporary** with an end date.
- Owner edits anyone; managers edit non-owners and cannot grant more than they have, pay rates or HR visibility.
- Every access change, and every owner action on another device, is logged.
- Money actions over limits (refunds, voids, discounts, stock write-offs) need a manager PIN.
- Devices are identities too: pings target devices; a lost device can be revoked.

5. Solution overview

A hub in the store holds all shared data and serves the app; every till, PC and phone connects to it over the staff Wi-Fi. Internet, from the router or a phone hotspot, is optional and only switches on connected services.

One hub on the shop Wi-Fi serves every device; internet is optional



system architecture · hub, network, devices, optional internet

Devices talk only to the hub. Heavy work (imports, analytics, bill reading) runs in the back-office or phone browser, so the Pi stays light.

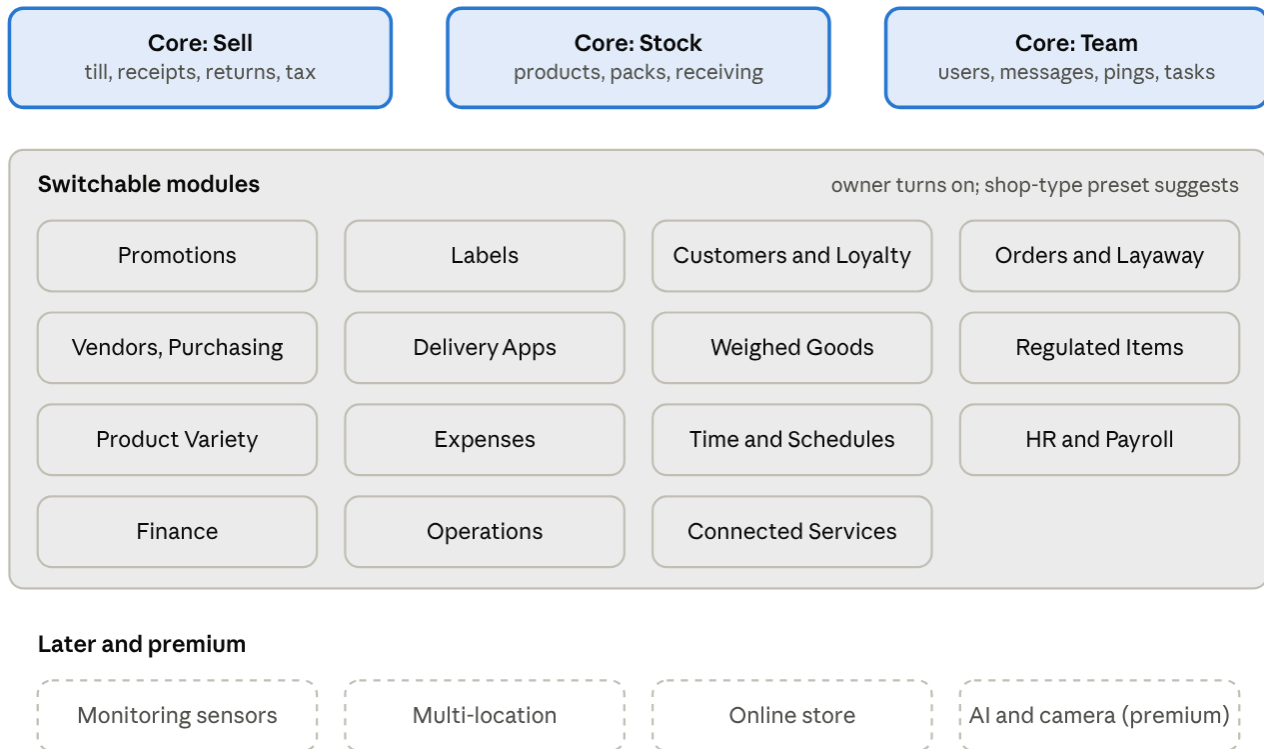
Connectivity modes

Mode	What works	What waits
Normal (LAN, no internet)	Everything in Core and switched-on modules; card terminal in its own offline/online mode	Remote view uploads, email fallback, updates, terminal link
Online (router or hotspot)	All of the above + connected services	—
Hub unreachable	Tills sell from cache and queue; phones keep drafts	Messages, pings, reports, stock sync until reconnect
Power cut	UPS keeps hub and router up; tills on battery continue	Clean shutdown on low battery

6. Module map and feature switches

Features are grouped into modules so each store sees only what it uses. Core is always on; the owner switches on other modules in setup (a shop-type preset pre-selects them) or later in Settings → Features.

Three core modules are always on; the owner switches on the rest



module map · 3 core, 15 switchable, 4 later

Switching a module off hides it from menus, screens, reports and the dashboard but never deletes its data.

Shop-type presets

Preset	Modules switched on
Grocery / perishables	Promotions, Labels, Customers and Loyalty, Vendors and Purchasing, Weighed Goods, Regulated Items, Operations
Convenience store	Promotions, Labels, Regulated Items, Vendors and Purchasing, Operations
Clothing / gifts	Promotions, Labels, Customers and Loyalty, Product Variety, Orders and Layaway

Preset	Modules switched on
Food takeout	Delivery Apps, Promotions, Customers and Loyalty, Operations
General retail	Promotions, Labels, Customers and Loyalty, Vendors and Purchasing

7. Functional requirements

Requirements are grouped by area. Columns: ID, requirement, module (Core or switchable), priority (M/S/C/L), phase.

FR-1 Setup, configuration, users and devices

ID	Requirement	Module	Pri	Phase
FR-1.01	First-start wizard: language, hub date/time check, business profile (names, type, address, GST/PST/BN numbers, currency, time zone, fiscal year)	Core	M	P0
FR-1.02	Branding: logo upload, colours (auto from logo), receipt header/footer, preview	Core	M	P0
FR-1.03	Owner account with password, PIN and printed recovery code	Core	M	P0
FR-1.04	Add managers and staff with role, PIN, contact; optional pay details	Core	M	P0
FR-1.05	Storage areas with temperature ranges (shelf, cooler, freezer, dry store)	Core	M	P0
FR-1.06	Shop-type preset that pre-selects modules; module on/off in Settings → Features (owner only); switching off hides but never deletes	Core	M	P0
FR-1.07	Device pairing by code/QR, including trusted-certificate install guidance for phones	Core	M	P0
FR-1.08	Device manager: online status, user, version; lock, log out, rename, assign, revoke, approve	Core	M	P0
FR-1.09	Owner can log in on any device	Core	M	P0

ID	Requirement	Module	Pri	Phase
FR-1.10	Role templates plus per-person permission overrides, optional end date; "Who can do this?" view	Core	M	P0
FR-1.11	Peripherals: find network receipt printer, test print, drawer kick; customer display	Core	M	P1
FR-1.12	Tax and payment method setup, till float default, cash rounding on/off	Core	M	P1
FR-1.13	Optional external references: bank, processor merchant ID, CRA payroll number, CBSA importer number, accountant	Core	S	P0
FR-1.14	Backup setup: USB drive, schedule, first backup; optional cloud drive copy	Core	M	P0
FR-1.15	Wizard resumable; skipped steps become dashboard tasks; re-runnable for edits, never for wiping	Core	M	P0
FR-1.16	Accessibility: large text, high contrast, screen-reader labels	Core	S	P1

FR-2 Dashboard, tasks, messages, pings, notifications

ID	Requirement	Module	Pri	Phase
FR-2.01	Role-based dashboard: attention list on top; KPI tiles and charts (sales, margin by category, P&L month to date, inventory health, wastage, payables/receivables)	Core	M	P2
FR-2.02	Rules engine creates tasks automatically (low/out of stock, expiring, expired, wrong storage, till variance, till not closed, refund approval, PO overdue, invoice due, expense/time sheet pending, missed punch, payroll due, margin drop, backup failed, hub health, import issues, setup incomplete, oversold while offline, Draft products) and closes them when fixed	Core	M	P2
FR-2.03	Manual tasks with owner, due date and link to a record	Core	S	P2

ID	Requirement	Module	Pri	Phase
FR-2.04	Messages: 1:1, groups, Everyone channel; attachments, mentions, reactions, read receipts, record links; follows the person across devices	Core	M	P2
FR-2.05	Pinned announcements needing acknowledgement; owner sees who acknowledged	Core	M	P2
FR-2.06	Pings: device-to-device, to groups (all tills, all phones), to all devices; templates; normal banner or urgent full-screen; reply in place; escalate if unacknowledged	Core	M	P2
FR-2.07	POS shows pings and messages as notifications over the sales screen without losing the sale	Core	M	P2
FR-2.08	In-app inbox for reports and alerts; email fallback only if unread by the deadline and the hub is online; at most one email per item (Section 8.4)	Connected Services	M	P2
FR-2.09	Per-type deadlines and subscriptions (end-of-day report, urgent alerts, messages); pings never emailed	Connected Services	S	P2
FR-2.10	End-of-day report delivered in-app to owner and chosen managers	Core	M	P2
FR-2.11	Opening/closing checklists and shift handover notes	Operations	S	P2
FR-2.12	Licence, permit and insurance expiry reminders	Operations	S	P2

FR-3 Checkout and payments

ID	Requirement	Module	Pri	Phase
FR-3.01	Till open with float by denomination; cash drops and pay-outs with reason; close with counted cash by denomination, expected vs counted, Z-report	Core	M	P1

ID	Requirement	Module	Pri	Phase
FR-3.02	Touch-first sales screen: category grid with product images, search, barcode scan (USB and phone camera), PLU entry	Core	M	P1
FR-3.03	Cart: quantity, line/cart discount (permission-limited), price override with reason and manager PIN above limit, notes, customer attach	Core	M	P1
FR-3.04	Weighed items: weight from certified scale or entered weight, unit kg/lb, tare; up to 3 decimals	Weighed Goods	M	P1
FR-3.05	Packs and singles: sells by selling unit; pack-break prompt when loose stock is 0	Core	M	P1
FR-3.06	Age-restricted items require "ID checked" confirmation, logged	Regulated Items	M	P1
FR-3.07	Container deposits and eco fees added automatically per item; deposit refunds	Regulated Items	M	P1
FR-3.08	Payments: cash (change calc, 5¢ rounding on cash only), card via standalone terminal (amount shown, approved/declined, last 4 or reference), store credit, USD cash at set rate with CAD change, platform payment for delivery orders	Core	M	P1
FR-3.09	Split payments across methods on one sale	Core	M	P1
FR-3.10	Hold a sale and recall it on any till	Core	M	P1
FR-3.11	Void line; void sale with manager PIN; no-sale drawer opens logged with reason	Core	M	P1
FR-3.12	Training mode: practice sales never affect stock, money or reports; clearly marked on screen and receipt	Core	M	P1
FR-3.13	Receipt: print via network printer with drawer kick; reprint; GST/HST receipt information by amount threshold; barcode for returns; loyalty summary; savings	Core	M	P1

ID	Requirement	Module	Pri	Phase
FR-3.14	Customer-facing display showing items, savings and total	Core	S	P2
FR-3.15	FEFO: sells oldest-expiry lot first; expired lots blocked without manager override	Core	M	P1
FR-3.16	Offline selling from cached catalogue with local queue; amber offline bar; upload on reconnect	Core	M	P1
FR-3.17	Hub processes stock and money actions one at a time; rejects lines that would go negative while online; soft hold on very low-stock items in carts (default ≤ 3 units, 5 min)	Core	M	P1
FR-3.18	Layaway and special orders: deposit, balance due, pickup	Orders & Layaway	S	P3
FR-3.19	Quotes converted to sale or invoice	Orders & Layaway	S	P3
FR-3.20	Store credit and house accounts with monthly statements	Customers & Loyalty	S	P3
FR-3.21	Linked card terminal (Square Terminal API first) when hub is online; auto fallback to standalone mode	Connected Services	S	P5
FR-3.22	Common payment interface for other processors (Helcim, Stripe Terminal, Moneris)	Connected Services	C	Later

FR-4 Tax, returns and exchanges

The tax and return rules below follow the v0.3 recommendation; they stay marked **pending accountant confirmation** until reviewed (Open question Q1).

ID	Requirement	Module	Pri	Phase
FR-4.01	Tax classes per product (Standard GST+PST, GST only, Zero-rated, Exempt, Custom)	Core	M	P1
FR-4.02	Tax rates in an updatable table with effective dates; provinces/HST as rows	Core	M	P1

ID	Requirement	Module	Pri	Phase
FR-4.03	Tax calculated per tax type per receipt, rounded once per type; discounts and loyalty spread across lines before tax	Core	M	P1
FR-4.04	Business setting: tax added at till, or tax-included prices	Core	M	P1
FR-4.05	Tax-exempt sales with reason (First Nations status card, PST resale/wholesale, other) and reference number; exempt report	Core	M	P1
FR-4.06	Every sale stores the tax it charged per line and per type; history never recalculated	Core	M	P1
FR-4.07	Return policy settings: window (per category), non-returnable flag, with/without receipt rules, refund limit without manager, optional restocking fee	Core	M	P1
FR-4.08	Find sale by receipt barcode, date, amount, card last 4 or customer	Core	M	P1
FR-4.09	Refund = original price paid after discount share + original tax; cannot return more than sold	Core	M	P1
FR-4.10	No-receipt returns: manager approval; store credit at lowest price in last 30 days	Core	M	P1
FR-4.11	Per returned item: back to stock, damaged, return to vendor, dispose	Core	M	P1
FR-4.12	Loyalty points reversed for returned items; redeemed points returned proportionally	Customers & Loyalty	M	P2
FR-4.13	Exchange on one screen (return + new sale, difference paid/refunded)	Core	M	P1
FR-4.14	Card refunds: on standalone terminal with cashier confirmation (Stage 1); via linked terminal (Stage 2)	Core / Connected	M	P1 / P5
FR-4.15	Tax reports by type and rate, laid out for GST/HST and BC PST returns	Finance	M	P4

FR-5 Products, pricing, promotions and labels

ID	Requirement	Module	Pri	Phase
FR-5.01	Product record: name, category, barcodes, PLU, unit, cost, price, live margin and markup, tax class, image, reorder point, perishable details, HS code and origin, bilingual name	Core	M	P1
FR-5.02	Product status Draft/Active; Active only when all required fields for its type pass validation (Section 9)	Core	M	P1
FR-5.03	Selling units: single, pack, case with conversions; nested; per-unit price, barcode and margin	Core	M	P1
FR-5.04	Autofill from bundled product data when a barcode is known; HS code, shelf life and storage suggested from name	Core	S	P1
FR-5.05	Price history for every price and cost change	Core	M	P1
FR-5.06	Scheduled prices with start/end date	Promotions	M	P2
FR-5.07	Promotion types: % off, \$ off, fixed price, buy Y get X% off, buy Y get \$X off, Y for \$X, buy Y get Z free, mix and match, spend threshold, time-based, near-expiry markdown, coupon	Promotions	M	P2
FR-5.08	Promotion scope: products, categories, exclusions; dates, days, hours; limits per transaction/customer/total	Promotions	M	P2
FR-5.09	Promotion preview: regular vs promo price, cost, new margin %; below-cost rows warned	Promotions	M	P2
FR-5.10	No stacking by default; best single deal; owner can allow stacking per promotion; priority item > category > cart	Promotions	M	P2
FR-5.11	Near-expiry markdown: automatic % by days left, applied per lot, label queued	Promotions	M	P2
FR-5.12	Promotion reports: units, sales, margin, savings vs prior period	Promotions	S	P2

ID	Requirement	Module	Pri	Phase
FR-5.13	Label batch: add by scan, category or list; auto-add on price change, promotion start/end, markdown, new product; duplicates merged	Labels	M	P1
FR-5.14	Label template: name, price, unit price (per kg/lb/100 g), barcode, promo text and end date, struck regular price, PLU, origin, bilingual name, logo	Labels	M	P1
FR-5.15	Layouts: Letter 3×10, Letter 2×5, A4 3×8, plain Letter/A4 with cut lines, custom rows × columns × margins, thermal roll	Labels	M	P1
FR-5.16	Start-at-position on partly used sheets; exact-size PDF; confirm printed; reprint last 10 batches; alignment test page	Labels	M	P1
FR-5.17	Variants (size × colour) under one parent product	Product Variety	S	P3
FR-5.18	Bundles/kits made of stocked items	Product Variety	S	P3
FR-5.19	Serial/IMEI capture at sale and warranty lookup	Product Variety	C	P3
FR-5.20	Staff discount rules with limits, logged	Core	S	P2

FR-6 Inventory, receiving and packs

ID	Requirement	Module	Pri	Phase
FR-6.01	Stock levels per product: on hand (sealed packs + loose units), reserved, available, incoming; by storage area and lot with expiry; colour status	Core	M	P1
FR-6.02	Add stock by phone: scan barcode → existing product page (add qty, cost, lot/expiry) or new-product form with barcode filled; type-code fallback; torch	Core	M	P1
FR-6.03	"Packs or singles?" prompt when one barcode is shared	Core	M	P1

ID	Requirement	Module	Pri	Phase
FR-6.04	Quick bulk add grid: scan per row, product, unit, cost per unit (last cost default, ▲/▼ %), quantity, lot/expiry; continuous scan; one-transaction save; drafts kept on phone	Core	M	P1
FR-6.05	Pack break at till (confirm) or from inventory; pack make (e.g. hampers); damaged units on opening	Core	M	P1
FR-6.06	Damage and loss entries with reason codes, optional photo, approval above value	Core	M	P1
FR-6.07	Shrink / loss report by reason, product, value	Core	M	P1
FR-6.08	Stock counts (full); variance review and approval	Core	M	P1
FR-6.09	Cycle counts by aisle/category with phone scanner	Core	L	Later
FR-6.10	Reserved client orders lower available stock	Orders & Layaway	S	P3
FR-6.11	Receive from vendor bill photo: read lines, match vendor and products (vendor code → barcode → fuzzy), review grid, totals and tax checks, PO check, confirm; learns matches	Vendors & Purchasing	M	P3
FR-6.12	Landed cost for imports (freight, duty, brokerage) spread across lines	Vendors & Purchasing	S	P3
FR-6.13	Min/max reorder rules generate draft POs	Vendors & Purchasing	S	P3
FR-6.14	Consignment stock: vendor-owned until sold; vendor payable on sale	Vendors & Purchasing	C	P3
FR-6.15	Recall trace: lots sold, when, to which loyalty customers; block recalled lots	Core	S	P3
FR-6.16	Wrong-storage alert when product temperature range doesn't match its storage area	Core	S	P2
FR-6.17	Monitoring menu: temperature sensors and logs per storage area	Monitoring	L	Later

ID	Requirement	Module	Pri	Phase
FR-6.18	Multi-location stock and transfers	—	L	Later

FR-7 Customers and loyalty

ID	Requirement	Module	Pri	Phase
FR-7.01	Customer profile: first name, phone and/or loyalty card number, purchase history, notes; never contacted by Chedam	Customers & Loyalty	M	P2
FR-7.02	Find customer at till by phone number or loyalty card scan; join in under 10 seconds with "customer agreed" tick	Customers & Loyalty	M	P2
FR-7.03	Owner sets earn rate, point value, minimum to redeem, exclusions, earn-on-promotions, expiry, bonus rules	Customers & Loyalty	M	P2
FR-7.04	Redeem points as discount before tax; receipt shows earned, redeemed, balance	Customers & Loyalty	M	P2
FR-7.05	Loyalty cards generated by Chedam, printable via label layouts; link card to phone; block lost card and move points	Customers & Loyalty	M	P2
FR-7.06	Offline earn/redeem against cached balance; double-use flagged as task	Customers & Loyalty	M	P2
FR-7.07	Loyalty reports: members, active, points liability in \$, top customers	Customers & Loyalty	S	P2
FR-7.08	Customer data access and deletion on request (BC PIPA)	Customers & Loyalty	M	P2
FR-7.09	Feedback portal: in-store tablet kiosk, or QR on receipt when the hub is online; rating + comment; feedback report	Customers & Loyalty	S	P3
FR-7.10	Customer pre-orders with arrival notice	—	L	Later

FR-8 Vendors, purchasing and delivery apps

ID	Requirement	Module	Pri	Phase
FR-8.01	Party records (vendor, client or both): contacts, addresses, payment terms, currency, tax ID, importer/exporter details, documents, notes	Vendors & Purchasing	M	P3
FR-8.02	Vendor products with vendor SKU and cost; supplier price list import; vendor comparison	Vendors & Purchasing	S	P3
FR-8.03	Purchase orders: draft, sent (PDF/print), partial/full receiving, differences flagged	Vendors & Purchasing	M	P3
FR-8.04	Bills (payables) and client invoices (receivables) with due dates, payments, statements	Vendors & Purchasing	M	P3
FR-8.05	Vendor returns and credits linked to PO	Vendors & Purchasing	M	P3
FR-8.06	Vendor performance: on-time %, short-shipped %, damage %	Vendors & Purchasing	C	P3
FR-8.07	Communication log and follow-up reminders per party	Vendors & Purchasing	S	P3
FR-8.08	Multi-currency documents with exchange rate on document date	Vendors & Purchasing	S	P3
FR-8.09	Delivery orders, manual mode: platform, order number, items, platform price per product; statuses Accepted (reserve) → Preparing → Picked up (deduct, sale) / Cancelled (release)	Delivery Apps	M	P3
FR-8.10	Platform payout statement import and commission tracking	Delivery Apps	S	P4
FR-8.11	Delivery orders automatic mode via aggregator or direct partner API when hub is online; menu availability sync	Delivery Apps	L	Later (partner track)

ID	Requirement	Module	Pri	Phase
FR-8.12	Snap-the-ticket: read platform ticket photo into an order	Delivery Apps	L	Premium

FR-9 Expenses, time, HR and payroll

ID	Requirement	Module	Pri	Phase
FR-9.01	Expense claim: date, vendor, amount, tax, category, payment, note, receipt photo; Submitted → Approved/Rejected → Reimbursed (till cash, cheque, e-transfer, next pay)	Expenses	M	P3
FR-9.02	Duplicate and missing-receipt flags	Expenses	M	P3
FR-9.03	Petty cash float with receipts and top-ups	Expenses	S	P3
FR-9.04	Clock in/out and breaks on any device with PIN; hub clock used for all times	Time & Schedules	M	P4
FR-9.05	Punch edits keep original and new time, who and why	Time & Schedules	M	P4
FR-9.06	BC break and overtime alerts before they happen (30-min meal break within 5 hours; overtime over 8 h/day or 40 h/week)	Time & Schedules	M	P4
FR-9.07	Weekly roster grid; publish; staff see own schedule; swap requests; labour cost vs sales	Time & Schedules	M	P4
FR-9.08	Time sheet approval per pay period with anomaly flags	Time & Schedules	M	P4
FR-9.09	Employee records: personal details, emergency contact, job, start date, pay type and rate, vacation entitlement, documents; SIN encrypted, owner-only	HR & Payroll	M	P4
FR-9.10	Leave balances (vacation, sick, unpaid) accrue automatically	HR & Payroll	S	P4

ID	Requirement	Module	Pri	Phase
FR-9.11	Payroll preparation: approved hours, overtime, vacation pay, reimbursements → gross; deductions entered from CRA calculator or accountant; line-by-line review; pay stubs	HR & Payroll	M	P4
FR-9.12	Pay stubs and statements visible to each employee; payment record (cheque number, deposit reference)	HR & Payroll	M	P4
FR-9.13	ROE data and T4 data ready for filing	HR & Payroll	M	P4
FR-9.14	Payroll service connection for automatic deductions and filing	HR & Payroll	L	Later
FR-9.15	Sales by employee and commissions	—	L	Later

FR-10 Finance, reconciliation and reports

ID	Requirement	Module	Pri	Phase
FR-10.01	Till reconciliation: each close matched to sales and card terminal settlement	Core	M	P1
FR-10.02	Bank reconciliation: import CSV/OFX/QFX; auto-match deposits, vendor payments, expenses, payroll; manual match; unmatched list	Finance	M	P4
FR-10.03	Bank deposit tracking: cash moved from tills to bank, matched on statement	Finance	M	P4
FR-10.04	Vendor statement reconciliation	Finance	S	P4
FR-10.05	Card processing fee tracking by card type; in margin and P&L	Finance	M	P4
FR-10.06	Period close and lock; owner unlock	Finance	M	P4
FR-10.07	Financial statements: profit & loss, balance sheet, cash flow	Finance	M	P4
FR-10.08	GST/HST and PST return reports (see FR-4.15)	Finance	M	P4
FR-10.09	6-year record retention: deletion blocked inside window; retention rules per record type	Core	M	P1

ID	Requirement	Module	Pri	Phase
FR-10.10	Sales insights: by hour/day heatmap, best/worst sellers, sell-through, year-over-year, margin by category	Core	M	P2
FR-10.11	Loss-prevention report: voids, refunds, no-sales, discounts, overrides per cashier with unusual patterns flagged	Core	M	P1
FR-10.12	Accountant exports: sales journal, expenses, payroll journal, reconciliation (QuickBooks/Xero-compatible CSV)	Finance	M	P4
FR-10.13	Audit log viewer: who changed what, when, on which device	Core	M	P1

FR-11 Import, export and bundled datasets

ID	Requirement	Module	Pri	Phase
FR-11.01	Import CSV, TSV, Excel, JSON, Parquet, zip bundles; bank CSV/OFX/QFX	Core	M	P1 (products) / P5 (all)
FR-11.02	Detect file type, encoding, delimiter, header row, date/number formats; profile columns with DuckDB in the back-office browser	Core	M	P1
FR-11.03	Identify target table; map columns by synonyms, fuzzy matching and value patterns with confidence levels	Core	M	P1
FR-11.04	Transform: units, currency, name splits, tax-inclusive to exclusive, dates; detect pack sizes in names	Core	M	P1
FR-11.05	Validate with the same rules as manual entry; failing rows become Drafts with reasons	Core	M	P1
FR-11.06	Gap report: missing tables, orphan references, duplicates, suspicious values; fix, bulk-fix, exclude or create missing records	Core	M	P1

ID	Requirement	Module	Pri	Phase
FR-11.07	Commit in one transaction with rollback; import job log	Core	M	P1
FR-11.08	Saved mapping templates (Square, Shopify, Lightspeed, Clover, Loyverse, QuickBooks, Xero, Wave, Zoho, generic)	Core	S	P5
FR-11.09	Export any list to CSV/Excel/PDF; full business export (CSV + JSON + data dictionary)	Core	M	P1
FR-11.10	Bundled datasets: HS codes, Canada tariff, India ITC-HS, Open Food Facts subset, Canadian Nutrient File, USDA FoodData, FoodKeeper/Handbook 66 shelf life, currency history, units, tax presets	Core	S	P5
FR-11.11	Reference data only suggests; never overwrites business data; licence credits in Settings → About	Core	M	P5

FR-12 Connected services and operations

ID	Requirement	Module	Pri	Phase
FR-12.01	Connectivity status: hub shows Offline / Online (router) / Online (hotspot); services pause and resume automatically	Connected Services	M	P0
FR-12.02	Update channel: when online, download signed packages (tax tables, app, schema, datasets, OS); install outside trading hours with backup and rollback	Core	M	P0
FR-12.03	Tax table updates switch on at their effective date; owner notified of what changes	Core	M	P0
FR-12.04	USB update package for stores that never connect	Core	M	P0

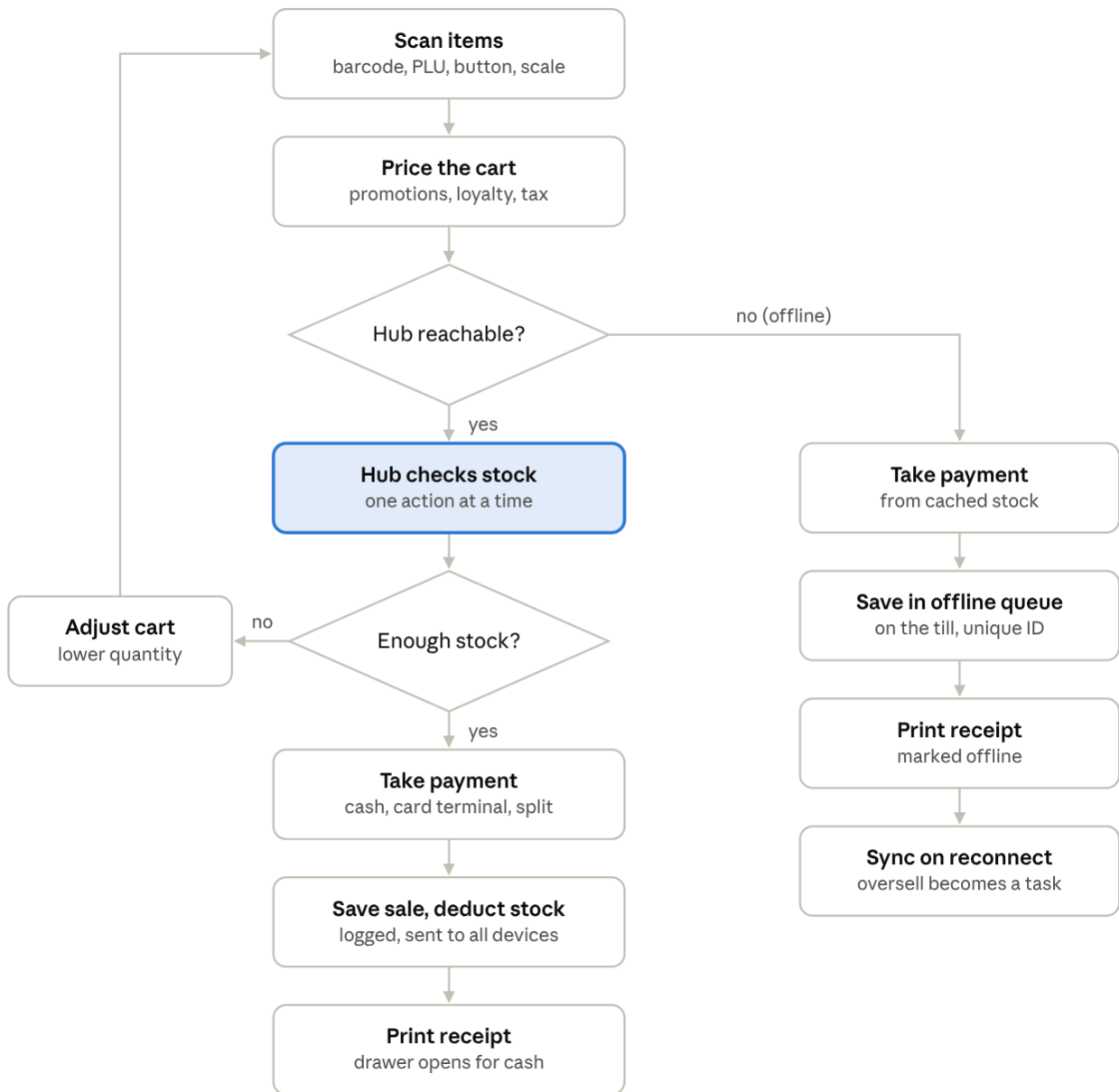
ID	Requirement	Module	Pri	Phase
FR-12.05	Owner remote view: owner links Google Drive or OneDrive (app-created folder only); hub uploads encrypted summary every 15 min (setting) and nightly backup; owner's app reads it remotely, read-only, with "last updated" time	Connected Services	M	P5
FR-12.06	Optional manager remote access with revocable key	Connected Services	C	P5
FR-12.07	Email fallback through owner's connected Gmail/Outlook account (FR-2.08)	Connected Services	M	P2
FR-12.08	Encrypted off-site backup to owner's cloud drive	Connected Services	M	P0
FR-12.09	Health page: uptime, temperature, RAM, disk, last backup, clock source, devices, queue backlog, versions, pending updates	Core	M	P0
FR-12.10	Remote support session, switched on by owner when needed	Connected Services	S	P5
FR-12.11	Built-in help and short training videos stored on the hub	Core	S	P2

8. Key process flows

8.1 Checkout

When the hub is reachable, stock is checked before payment so nothing is oversold; when it is not, the till keeps selling and reconciles later.

Stock is checked before payment whenever the hub is reachable



8.1 checkout · online and offline paths

Related: FR-3, BR-10 to BR-16.

8.2 Offline sync

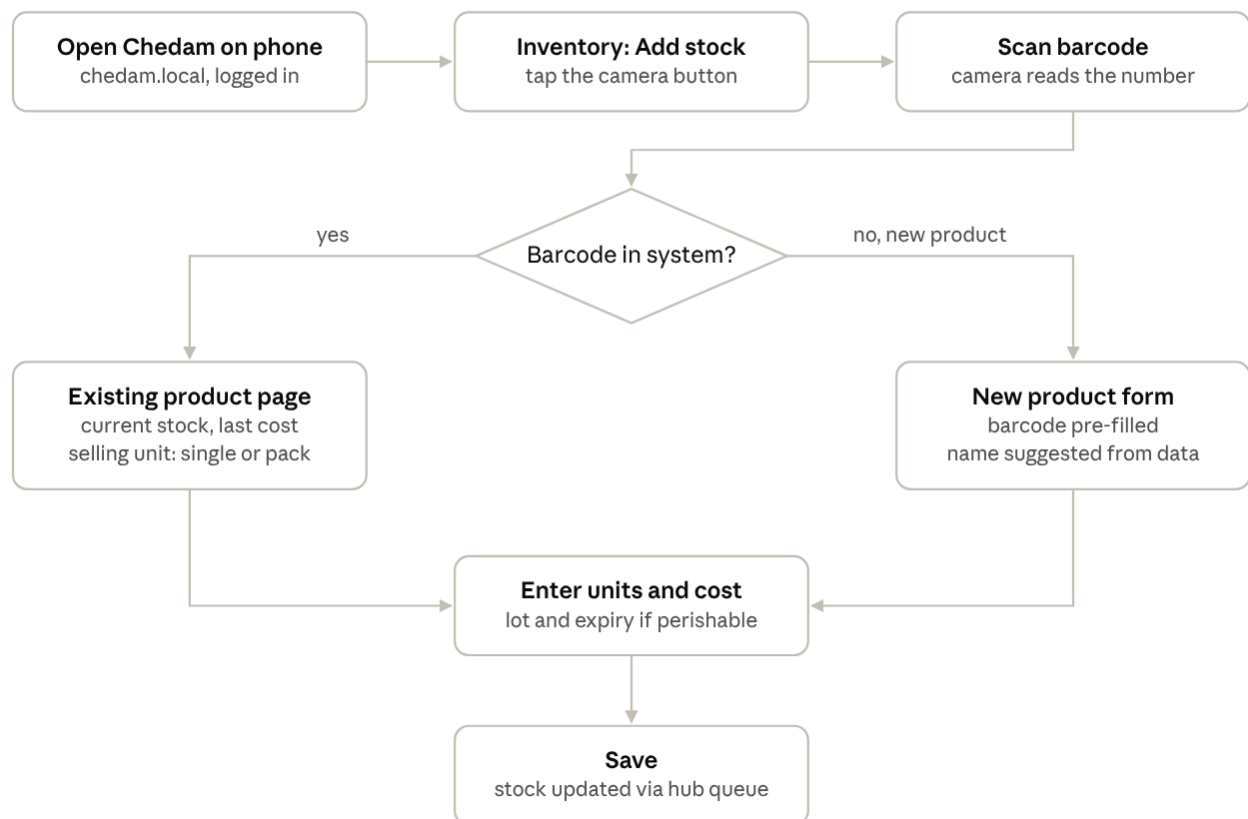
1. Till detects the hub is unreachable → amber "Offline" bar; sells from cached catalogue, prices, promotions and loyalty balances.
2. Each sale, return and till action is saved locally with a unique ID, device ID and device time.
3. Every few seconds the till retries the hub.

4. On reconnect the queue uploads in order; the hub processes each through the same one-at-a-time queue, using its own clock for official times and keeping the device time for reference.
5. Duplicate IDs are ignored (result returned); oversold stock is accepted as negative and an urgent task is raised; loyalty double-use is flagged.
6. The till downloads everything it missed (prices, stock, messages) and the bar turns green.

Related: FR-3.16, BR-10 to BR-12.

8.3 Add stock by phone

Every scan lands on a product page, existing or new

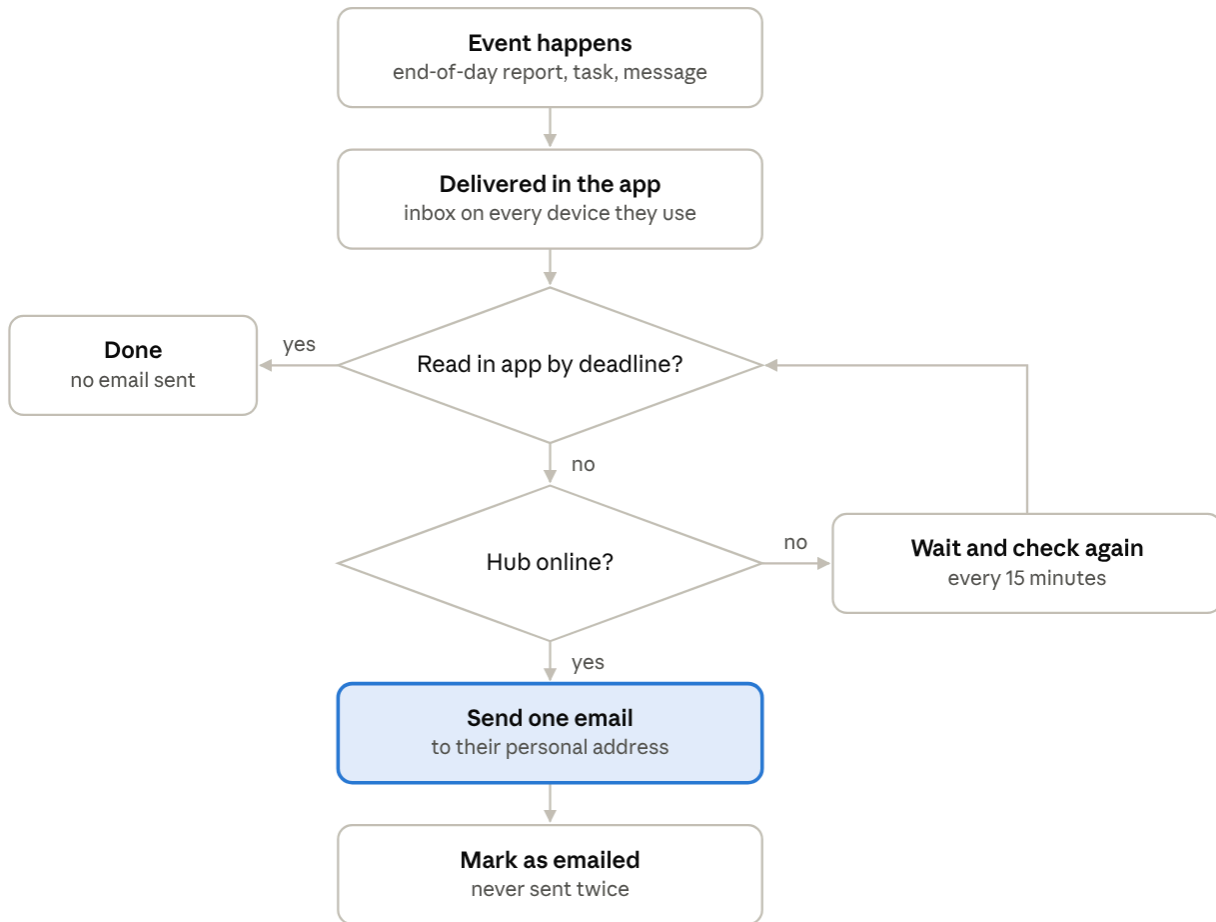


8.3 add stock by phone · scan, match, save

Related: FR-6.02 to FR-6.04.

8.4 Notification delivery

Email goes out only when a message is unread and the hub is online



8.4 notification delivery · in-app first, email fallback

Related: FR-2.08, FR-2.09, BR-40 to BR-42.

9. Business rules and validation catalogue

Business rules have IDs (BR-x) so code and tests can cite them. Defaults marked * are owner-editable settings.

Data validation

ID	Rule
BR-01	Money stored as integer cents; displayed with 2 decimals; no negatives; sellable price $\geq$ \$0.01
BR-02	Weight up to 3 decimals, > 0 on sales; no negatives

ID	Rule
BR-03	"Each" quantities are whole numbers > 0 on sales and receiving; weighed quantities up to 3 decimals
BR-04	Percentages 0-100, up to 3 decimals
BR-05	Negative stock adjustments only with a reason code
BR-06	1 lb = 0.45359237 kg; rounding only for display
BR-07	Product becomes Active only when required fields for its type pass: all products (name, category, selling unit, price, tax class, barcode/PLU/button); by weight (unit, price per unit, PLU/scale code, tare, approved scale acknowledged); pack (contains qty + unit); perishable (shelf life or expiry at receiving, storage area); age-restricted (minimum age); deposit/fee (amount, tax treatment); imported (HS code, origin)
BR-08	Save as Draft always allowed; Drafts cannot be sold; a task lists Drafts

Selling and stock

ID	Rule
BR-10	Hub processes stock and money actions one at a time in a single transaction; duplicate IDs return the original result
BR-11	Online: a sale that would make stock negative is rejected before payment
BR-12	Offline: sales queue locally; on sync, oversells are accepted, stock shows the true negative, and an urgent task is raised
BR-13	Soft hold on items with $\leq 3^*$ units when added to a cart, released after 5^* min, on sale or removal
BR-14	FEFO: oldest expiry lot sold first; expired lots blocked without manager override

ID	Rule
BR-15	Pack break requires confirmation and records person, device, time
BR-16	Cash total rounds to nearest 5¢ after tax; card totals exact
BR-17	Refund over \$50* or without receipt needs manager PIN
BR-18	Void sale, price override above limit*, stock write-off above \$ value* need manager PIN

Pricing, tax and loyalty

ID	Rule
BR-20	One promotion per item unless marked stackable; best price for customer; priority item > category > cart
BR-21	Cart discounts and loyalty redemptions spread across lines by price before tax
BR-22	Tax per type per receipt, rounded once per type
BR-23	Returns refund the original price paid after discount share plus original tax; quantity returned ≤ quantity sold
BR-24	Loyalty earn = 1* point per \$1 after discounts, before tax; 100* points = \$1; minimum 500* to redeem
BR-25	Price, cost or promotion change auto-adds the product to the label batch

People and time

ID	Rule
BR-30	All times from the hub clock, stored in UTC, shown in store time zone

ID	Rule
BR-31	BC: warn before 5 consecutive hours without a 30-min meal break; overtime after 8 h/day or 40 h/week
BR-32	Edits to punches keep original values, editor and reason
BR-33	Managers cannot grant permissions they do not hold, nor see pay rates
BR-34	Records retained at least 6 years; deletion inside window blocked

Notifications

ID	Rule
BR-40	Every notification goes to the in-app inbox first
BR-41	Email only if unread at deadline and hub online; one email per item, ever
BR-42	Pings are never emailed; urgent pings repeat until acknowledged; unacknowledged after 5* min escalate to managers

10. Data model

One SQLite database on the hub holds all live data; every table has `id` (created on the device), `created_at`, `updated_at`, `created_by`, `device_id`, `deleted_at` (soft delete). Money in integer cents, times in UTC.

Area	Tables (key fields)
Business	<code>business</code> (names, logo, colours, address, tax numbers, currency, time zone, fiscal year, tax display mode) · <code>location</code> · <code>settings</code> (key/value) · <code>modules</code> (module, enabled, enabled_by/at)

Area	Tables (key fields)
People and access	users (name, contact, role, PIN hash, status) · roles · permissions · permission_overrides (user, permission, allow/deny, expires_at, granted_by) · devices (name, type, key, last seen, assigned printer, revoked)
Catalogue	products (status Draft/Active, name, bilingual name, category, base unit, tax class, perishable data, HS code, origin, age limit, image, reorder point) · selling_units (product, name, barcodes, contains qty/unit, price, sell at POS) · categories · price_history · deposits_fees · variants · bundles
Tax	tax_types (GST, PST, HST) · tax_rates (type, province, rate, effective from/to) · tax_classes (which types apply) · tax_exemptions (sale, reason, reference)
Promotions and labels	promotions (type, X, Y, Z, scope, dates, days/hours, limits, stackable, priority) · promotion_scope · coupons · label_batch_items (product, unit, qty, reason, added_at) · label_templates · label_layouts (paper, columns, rows, margins, gaps)
Stock	storage_areas (name, temp range) · stock_levels (product, sealed per unit, loose, reserved, incoming) · stock_lots (lot, expiry, qty, cost) · stock_movements (type: sale, receive, adjust, damage, loss, return, transfer, pack_break, pack_make; qty; reason; reference) · stock_counts
Sales	tills (float, open/close, expected vs counted, drops, pay-outs) · sales (number, till, cashier, customer, totals, status, training flag, offline flag) · sale_lines (product, unit, lot, qty, price, discount share, promotion, tax per type, cost at sale) · payments (method, amount, processor, reference, last4, status) · returns / return_lines · holds · soft_holds · layaways · quotes

Area	Tables (key fields)
Customers	customers (first name, phone, card number, consent tick, notes) · loyalty_accounts (points balance) · loyalty_ledger (earn/redeem/reverse, sale) · store_credit_ledger · feedback
Parties and purchasing	parties (vendor/client) · party_contacts · vendor_products (vendor SKU, cost, learned match) · purchase_orders / po_lines · bills / invoices (+ lines, payments, attachments) · bill_scans (image, extracted lines, matches, status) · delivery_orders (platform, order no., status, payout, commission)
Team	conversations / messages (read receipts) · pings (from, to device/group, priority, acknowledged) · notifications (type, recipient, deadline, read_at, emailed_at) · tasks · checklists / checklist_runs · handover_notes · reminders
Money	expenses · petty_cash · bank_accounts / bank_lines · deposits · reconciliations · periods (locked)
Time and pay	shifts · time_entries (in/out, breaks, original values, edits) · employees (pay, SIN encrypted) · leave · pay_runs / pay_stubs
System	events (audit/change log: who, device, table, record, before/after, time) · import_jobs · updates (package, version, status) · remote_sync (provider, last upload) · backups

Event log: every create, update and delete writes an `events` row. It powers the audit trail, offline replay, undo, history charts and future AI features.

11. Non-functional requirements

ID	Category	Requirement	Target
NFR-01	Availability	Tills keep selling with hub or internet down	100% of cash sales offline

ID	Category	Requirement	Target
NFR-02	Reliability	No lost or duplicated sales	0 in pilot
NFR-03	Recovery	Restore a failed hub from backup	≤ 15 min with spare card
NFR-04	Backup	Nightly verified backup; 14 daily, 8 weekly, 12 monthly; optional cloud copy	100% nights logged
NFR-05	Performance	Scan-to-cart and payment-to-receipt on till	< 1 s on pilot hardware
NFR-06	Performance	Stock change visible on other devices	≤ 2 s
NFR-07	Capacity	Pilot hub (Pi Zero 2 W) devices	5 devices; production hub 10–30
NFR-08	Resources	Hub app memory	< 150 MB on Pi Zero 2 W
NFR-09	Power safety	Survives power pull mid-sale without corruption	Tested every release
NFR-10	Time	Correct time without internet (RTC)	Drift < 2 s/day
NFR-11	Security	HTTPS on LAN; hub-side permission checks; PIN lockout after 5 tries; auto-lock	Every release
NFR-12	Security	Sensitive fields (SIN, bank, pay rates) and backups encrypted	Always
NFR-13	Privacy	BC PIPA: minimum personal data; access/deletion on request; retention rules	Always
NFR-14	Compliance	GST/HST receipt information; 5¢ cash rounding; 6-year retention; Measurement Canada-approved scales for weight pricing	Before pilot
NFR-15	Usability	New cashier sells unaided after training	≤ 30 min training

ID	Category	Requirement	Target
NFR-16	Usability	Touch targets on POS	≥ 48 px; large text option
NFR-17	Accessibility	Contrast and screen-reader labels	WCAG 2.1 AA on core screens
NFR-18	Browsers	Supported	Current Chrome, Edge, Safari, Firefox; iOS and Android phones
NFR-19	Portability	Same software on Pi Zero 2 W, Pi 4/5, mini PC	One build
NFR-20	Maintainability	Schema migrations versioned, reversible; update rollback automatic	Every release
NFR-21	Cost	Software running cost to the store	\$0/month
NFR-22	Data ownership	Full export available any time	CSV + JSON + dictionary

12. Hardware and software bill of materials

Hardware

Item	Development	Pilot store	Production store
Hub	Pi Zero 2 W (owned)	Pi Zero 2 W	Pi 4/5 (2–4 GB) or mini PC
Storage	SD card (owned)	High-endurance microSD 64 GB + spare	USB SSD 128–256 GB + spare
Power	Power cable (owned)	Official supply + small UPS	Official supply + UPS for hub and router
Clock	Not needed (home internet)	DS3231 RTC	DS3231 (Pi 4) / battery (Pi 5)
Network	Home Wi-Fi 2.4 GHz	USB OTG hub + USB-Ethernet	Built-in Ethernet
Backup	—	USB drive	USB drive + cloud copy

Item	Development	Pilot store	Production store
Display (setup)	Micro HDMI (owned)	—	—
POS peripherals	—	USB barcode scanner, network thermal printer, cash drawer	Same + certified scale, label printer (optional), customer display (optional)
Card terminal	—	Store's own Square (or similar)	Same; linked in P5

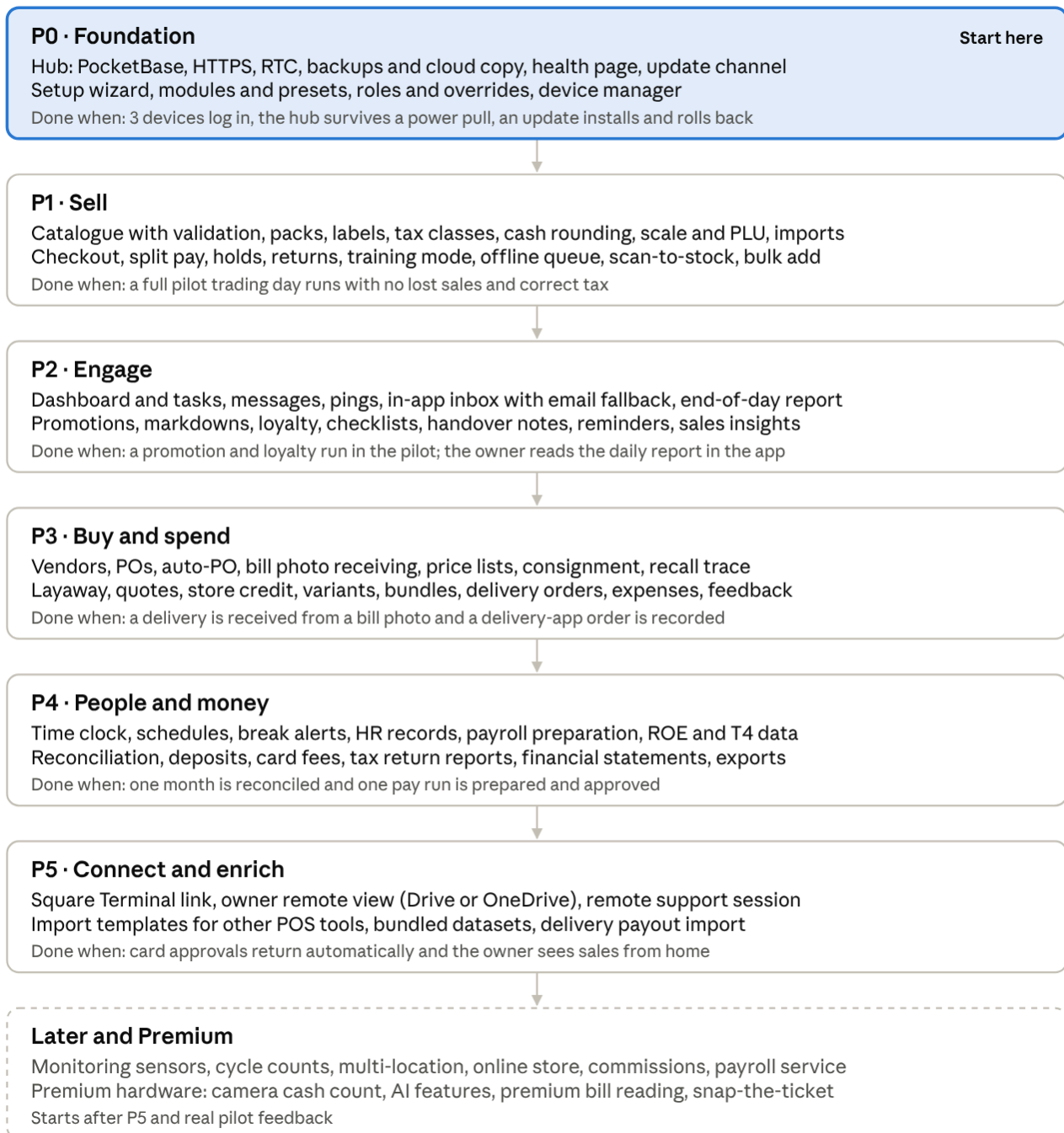
Software (all free / open source)

Layer	Component
Hub OS	Raspberry Pi OS Lite 64-bit
App server + database	PocketBase (Go) with SQLite (WAL mode)
HTTPS / local names	Caddy or PocketBase TLS with local CA (mkcert); Avahi mDNS (<code>chedam.local</code>)
Time / reliability	chrony + RTC; systemd restart; hardware watchdog
Client app	Svelte PWA, Tailwind CSS, Apache ECharts, Dexie.js (IndexedDB), service worker
Scanning and reading	Browser BarcodeDetector + ZXing; Tesseract.js for bill reading
Data import / analytics	DuckDB-WASM, SheetJS, Papa Parse
Printing	ESC/POS via hub; PDF generation in browser for labels and reports
Connected services	Google Drive / OneDrive APIs; Gmail / Outlook sending; Square Terminal API
Update channel	Signed packages on a free static host (e.g. GitHub Releases)
Development	VS Code, Git, Node.js + Vite, PocketBase on Windows, Raspberry Pi Imager, SSH, DB Browser for SQLite, DuckDB CLI

13. Release plan

Chedam ships in six phases. Each ends with a working release tested on the Pi and used in the pilot store before the next phase starts.

Six phases, each usable in the pilot store before the next starts



release plan • P0 to P5, then later and premium

P2 to P5 can be reordered if the pilot store needs something sooner; P0 and P1 cannot.

Phase gate (every phase)

- ☐ All Must requirements for the phase pass their tests
- ☐ Works offline where specified; permissions enforced on the hub
- ☐ Every write appears in the event log; relevant tasks create and close
- ☐ Exportable to CSV/Excel
- ☐ Tested on the Pi with 3 devices and a power pull mid-sale
- ☐ Backup and restore verified
- ☐ Pilot store has used it for at least one week
- ☐ Decision log and this specification updated

Build process per phase

1. Half-page mini-spec referencing requirement IDs.
2. Schema migrations and sample data first.
3. Build hub rules and client screens.
4. Test on the Pi; pilot; fix; tag release; publish update package.

14. Risks, assumptions, issues and dependencies

Risks

ID	Risk	Likelihood	Impact	Mitigation
R1	Owners won't switch without integrated card payments	High	High	Clear messaging; Stage 1 standalone; Stage 2 Square link in P5
R2	SD card or hub failure in 24/7 use	Medium	High	High-endurance card, UPS, backups, spare card, SSD in production
R3	Tax or receipt rules implemented wrongly	Medium	High	Accountant review before P1 release; tax tables updatable
R4	Payroll errors	Medium	High	Deductions entered from CRA calculator; payroll service later
R5	Phones block camera without trusted HTTPS	Medium	Medium	Guided certificate install; test iOS/Android in P0

ID	Risk	Likelihood	Impact	Mitigation
R6	Pi Zero 2 W too slow at peak	Medium	Medium	Load test in P0; production hardware
R7	Scope too large for one builder	High	High	Strict phases; module switches; each phase usable alone
R8	Delivery platform access refused	High	Medium	Manual mode; aggregator route later
R9	Bill photo reading inaccurate	Medium	Low	Review grid; learning; premium OCR
R10	Support load grows with each store	High	Medium	Health page, remote support (owner-enabled), help videos, support plan
R11	Liability for data loss or wrong totals	Low	High	Terms of service, warranty statement, insurance (to decide)

Assumptions

A1 Single location per hub · A2 BC first, Canada-wide via tax tables · A3 Store has Wi-Fi router · A4 Pilot store available (Amramphalam or friendly shop) · A5 Sreya provides support.

Issues (open now)

I1 Tax treatment of deposits/eco fees not confirmed · I2 Support plan and pricing model not defined (Gap list G).

Dependencies

D1 Square developer account and Terminal for P5 · D2 Google/Microsoft API app registration for remote view and email · D3 Measurement Canada-approved scale for weighed goods · D4 Accountant review · D5 Free hosting for update channel.

15. Decision log

Date	ID	Decision
2026-10-02	DL-20	Master Specification v1.0 baseline created from v0.1–v0.3 and gap-list decisions
2026-10-02	DL-19	Features grouped into switchable modules; shop-type presets
2026-10-02	DL-18	Labels: Letter and A4, user picks; rows × columns configurable
2026-10-02	DL-17	Remote view via owner's Google Drive or OneDrive, owner picks
2026-10-02	DL-16	Loyalty customers identified by phone number or printed card; never contacted
2026-10-02	DL-15	Notifications in-app first; email only if unread and hub online, once
2026-10-02	DL-14	Card payments: standalone Square-type terminal now; Terminal API later
2026-10-02	DL-13	Updates (tax, app, schema, datasets) when online; USB fallback
2026-10-02	DL-12	Temperature monitoring deferred to a later Monitoring menu
2026-10-02	DL-11	Gap-list Add/Later/Skip decisions adopted (see v0.3 Section 14)
2026-10-02	DL-10	Delivery apps: manual mode first; automatic when partner access exists
2026-10-02	DL-09	Offline tills keep selling; conflicts flagged as tasks
2026-10-02	DL-08	Owner logs in on any device, manages all devices, sends to all
2026-10-02	DL-07	Camera cash estimate in premium tier
2026-10-01	DL-06	AI is a later premium tier; foundations built now
2026-10-01	DL-05	One app for many businesses via setup wizard
2026-10-01	DL-04	Reference datasets bundled; no user downloads

Date	ID	Decision
2026-10-01	DL-03	DuckDB for import and analytics in the back-office browser
2026-10-01	DL-02	Hub: PocketBase + SQLite; client: PWA
2026-10-01	DL-01	Offline LAN architecture with hub; pilot on Pi Zero 2 W

16. Open questions

- ☐ Q1 Tax classes, zero-rated items, deposit/eco-fee tax treatment — accountant to confirm (blocks P1 release).
- ☐ Q2 Return policy defaults (30 days, \$50 cashier limit, store credit without receipt) — accept?
- ☐ Q3 First processor for Stage 2: Square, or compare Helcim/Stripe on cost?
- ☐ Q4 Remote view upload frequency: 15 minutes or end of day only?
- ☐ Q5 Email fallback deadlines (default next morning 8 am for end-of-day report).
- ☐ Q6 Pilot loyalty settings (earn rate, point value).
- ☐ Q7 Pack and single: shared or separate barcodes in the pilot store's products?
- ☐ Q8 Support plan, pricing model, terms and insurance (Gap list G).
- ☐ Q9 Pilot store and pilot start date.

17. Glossary

Term	Meaning
Hub	The small computer (Raspberry Pi) in the store that runs Chedam's server and database
PWA	Progressive Web App: a website that installs like an app and works offline
Offline queue	Sales and changes saved on a device while it cannot reach the hub, sent later
FEFO	First expiry, first out: sell the lot that expires soonest first
Lot	A batch of one product received together, with its own expiry date and cost

Term	Meaning
Selling unit	A way a product is sold: single, pack, case
Pack break	Opening a sealed pack so its units can be sold singly
PLU	Price look-up code for produce (e.g. 4011 bananas)
Z-report	End-of-day till report of sales, payments, refunds and voids
Tax class	Group of products taxed the same way (e.g. GST only)
Soft hold	Temporary reservation of a low-stock item in an open cart
Module	A group of features the owner can switch on or off
Event log	Record of every change: who, what, when, which device
MoSCoW	Must / Should / Could / Later priority labels
Stage 1 / Stage 2 payments	Standalone terminal with manual amount / terminal linked through the internet

18. Related documents and sources

Related Chedam documents

- [v0.1 Product & Technical Specification](#)
- [v0.2 Change Requests](#)
- [Reality Check and Retail Gap List](#)
- [v0.3 Decisions and New Features](#)

External sources

- [Square Terminal API — take payments](#)
- [Uber Eats Marketplace API — getting started](#)
- [DoorDash Marketplace integration requirements](#)
- [Cuboh — SkipTheDishes integration](#)
- [GST/HST invoice requirements — MapleInvoice](#)
- [Square Canada review — PayDaddy](#)

- [Loyverse POS review — POSUSA](#)

Rules cited from general knowledge (5¢ cash rounding, Measurement Canada scale approval, BC deposits, BC meal breaks and overtime, 6-year CRA retention, BC tax classes) must be confirmed with an accountant or the relevant agency before release.